

$$1. \quad E(k) [1 + \alpha E(k)] = \frac{\hbar^2 k^2}{2m_0 m_e}$$

$$(1) \quad E(k) \ll \frac{1}{\alpha} \quad (\Rightarrow) \quad \alpha E(k) \ll 1 \quad \Rightarrow \quad E(k) = \frac{\hbar^2 k^2}{2m_0 m_e}$$

$$(2) \quad E(k) \gg \frac{1}{\alpha} \quad (\Rightarrow) \quad \alpha E(k) \gg 1 \quad \Rightarrow \quad \alpha E^2(k) = \frac{\hbar^2 k^2}{2m_0 m_e} \quad \Rightarrow \quad E(k) = \frac{\hbar k}{\sqrt{2\alpha m_0 m_e}}$$

$$W = \frac{E}{\hbar} \quad (1) \quad V = \frac{dW}{dk} = \frac{\partial}{\partial k} \left(\frac{\hbar^2 k^2}{2m_0 m_e} \right) = \frac{2\hbar k}{2m_0 m_e} = \frac{\hbar k}{m_0 m_e}$$

$$(2) \quad v = \frac{\partial}{\partial k} \left(k \cdot \frac{1}{\sqrt{2\alpha m_0 m_e}} \right) = \frac{1}{\sqrt{2\alpha m_0 m_e}}$$

$$(1) \quad n_{3D}(E) = \frac{m}{\pi^2 \hbar^3} \sqrt{2mE} = \frac{2}{\pi^2 \hbar^3} \sqrt{2mE} = \frac{2}{\pi^2 \hbar^3} \sqrt{2mE}$$

$$3. \quad n_{3D}(E) = \frac{m}{\pi^2 \hbar^3} \sqrt{2mE} = \frac{1}{2\pi^2} \left(\frac{2m_0 m_e}{\hbar^2} \right)^{3/2} \sqrt{E}$$

$$N_{3D} = \int_0^{\infty} n_{3D}(E) dE = \frac{1}{2\pi^2} \left(\frac{2m_0 m_e}{\hbar^2} \right)^{3/2} \int_0^{\infty} \sqrt{E} \cdot \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}} dE = \frac{1}{2\pi^2} \left(\frac{2m_0 m_e}{\hbar^2} \right)^{3/2} \int_0^{\infty} \sqrt{E} \cdot e^{-\frac{E - E_F}{k_B T}} dE$$

$$u = \frac{E}{k_B T} \quad \Rightarrow \quad du = \frac{1}{k_B T} dE$$

$$\int_0^{\infty} \sqrt{E} \cdot e^{-\frac{E - E_F}{k_B T}} dE = \int_0^{\infty} \sqrt{u} \cdot e^{-u} \cdot k_B T du = \frac{1}{2\pi^2} \left(\frac{2m_0 m_e}{\hbar^2} \right)^{3/2} (k_B T)^{3/2} \cdot \frac{1}{2} \pi^{1/2}$$

$$= \frac{\pi^{1/2}}{2\pi^2} \cdot (2)^{3/2} \left(\frac{m_0 m_e}{\hbar^2} \right)^{3/2} (k_B T)^{3/2} = 2 \left(\frac{m_0 m_e k_B T}{\hbar^2 2\pi} \right)^{3/2}$$

$$4. \quad n_p = n_i^2 \quad (\Rightarrow) \quad p = \frac{n_i^2}{n}$$

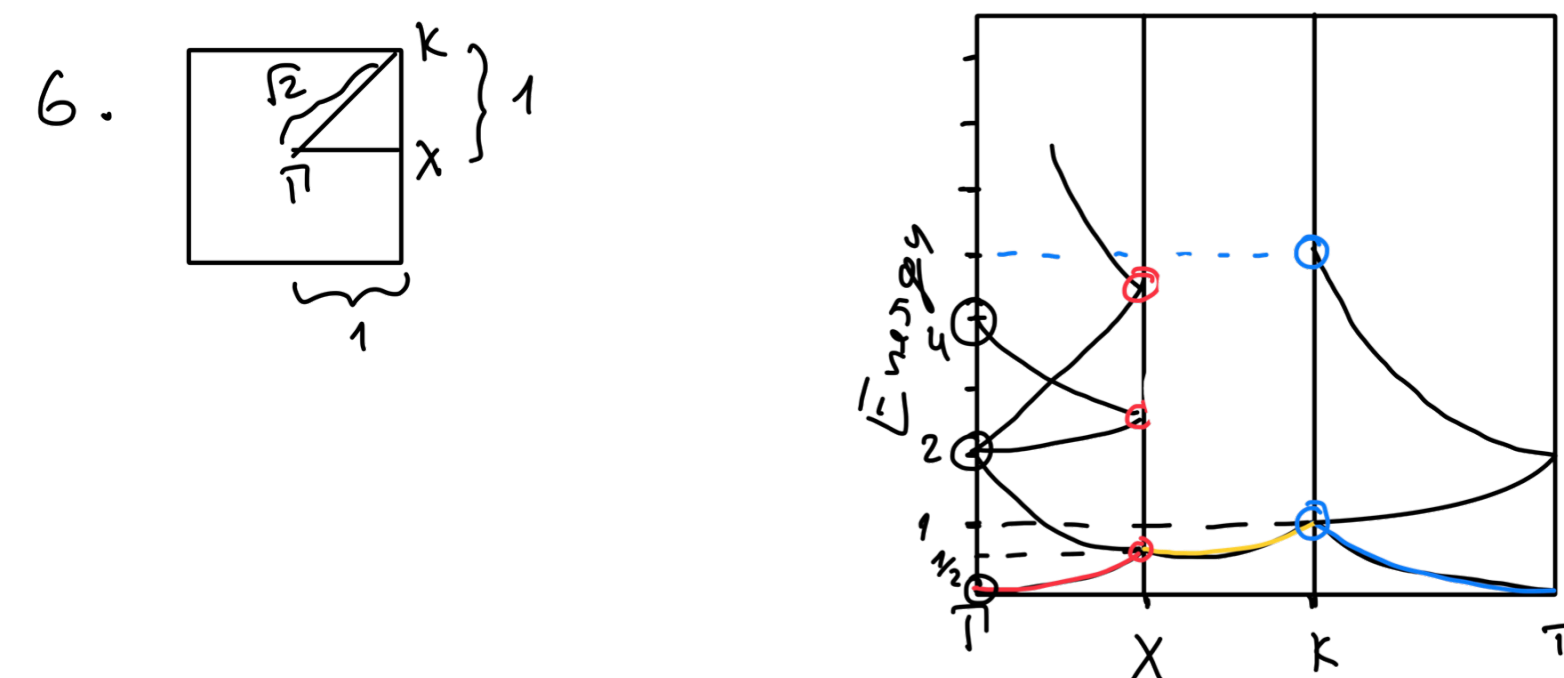
$$p + N_D = n + N_A \quad (\Rightarrow) \quad \frac{n_i^2}{n} + N_D = n + N_A \quad (\Rightarrow) \quad n + N_A - N_D = \frac{n_i^2}{n} \quad (\Rightarrow) \quad n_i^2 = n^2 + n(N_A - N_D) - n_i^2 = 0 \quad (\Rightarrow) \quad n = \frac{N_D - N_A \pm \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2} \quad \Rightarrow \quad n = \frac{1}{2} \left[N_D - N_A \pm \sqrt{(N_D - N_A)^2 + 4n_i^2} \right]$$

$$5. \quad \psi_k(x) = u_k(x) e^{ikx} \quad u_k(x+a) = u_k(x)$$

$$\psi_k(x+a) = u_k(x+a) e^{ik(x+a)} = u_k(x+a) e^{ikx} e^{ika}$$

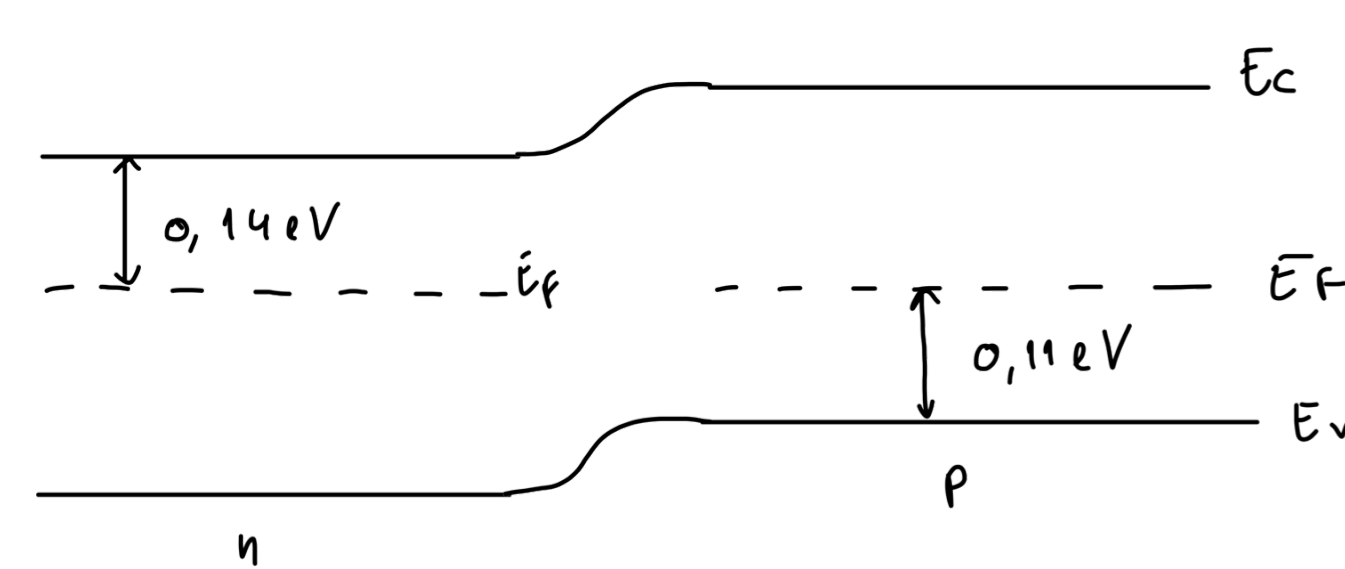
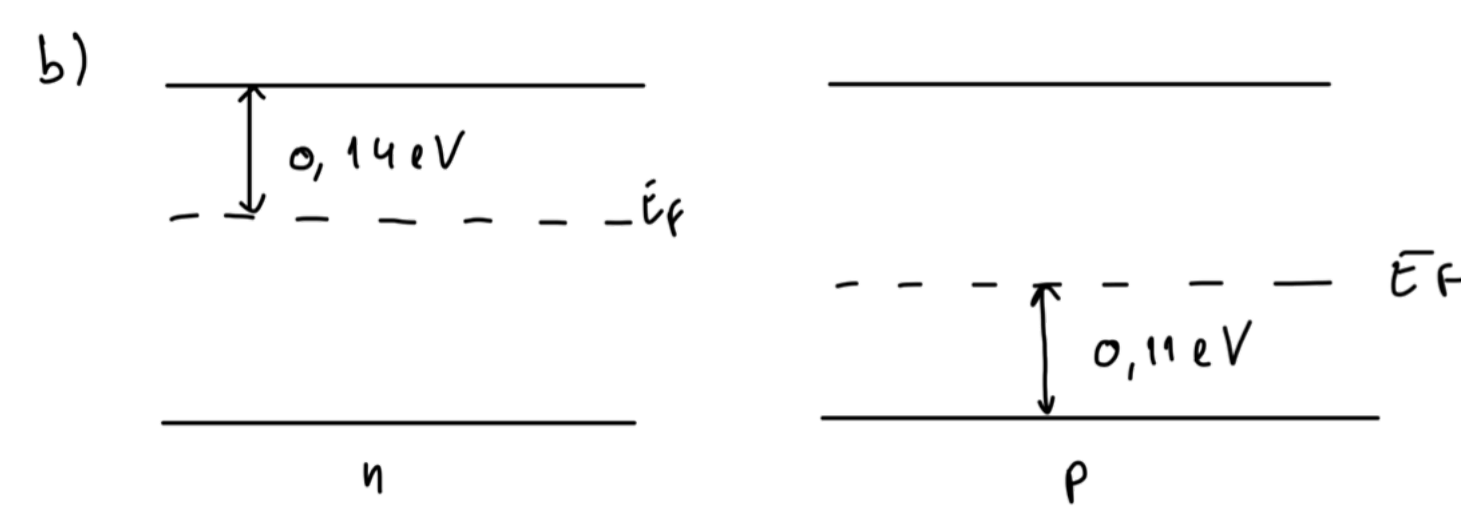
$$\psi_k(x+a) = u_k(x) e^{ikx} e^{ika}$$

$$\psi_k(x+a) = \psi_k(x) e^{ika}$$



$$7. \quad a) \quad n = N_C e^{-\frac{E_C - E_F}{k_B T}} \quad \text{bande de conduction} : n \approx N_D = N_C e^{-\frac{E_C - E_F}{k_B T}} \quad (\Rightarrow) \quad \frac{E_F - E_C}{k_B T} = \ln \left(\frac{N_D}{N_C} \right) \quad (\Rightarrow) \quad E_F - E_C = -k_B T \ln \left(\frac{N_D}{N_C} \right) \quad (\Rightarrow) \quad E_C - E_F = k_B T \ln \left(\frac{N_C}{N_D} \right) \quad (\Rightarrow) \quad E_C - E_F = 2,26 \times 10^{-20} \text{ J} = 0,14 \text{ eV}$$

$$p = N_V e^{-\frac{E_F - E_V}{k_B T}} = N_A \quad (\Rightarrow) \quad \frac{E_F - E_V}{k_B T} = \ln \left(\frac{N_A}{N_V} \right) \quad (\Rightarrow) \quad E_F - E_V = k_B T \ln \left(\frac{N_A}{N_V} \right) = 1,76 \times 10^{-20} \text{ J} = 0,11 \text{ eV}$$



$$c) \quad E_C - E_F = k_B T \ln \left(\frac{N_C}{N_D} \right) \quad (\Rightarrow) \quad E_F = E_C - k_B T \ln \left(\frac{N_C}{N_D} \right)$$

$$E_F - E_V = k_B T \ln \left(\frac{N_V}{N_A} \right) \quad (\Rightarrow) \quad E_F = E_V + k_B T \ln \left(\frac{N_V}{N_A} \right)$$

$$V_{bi} = E_F^n - E_F^p = E_C - k_B T \ln \left(\frac{N_C}{N_D} \right) - \left(E_V + k_B T \ln \left(\frac{N_V}{N_A} \right) \right) = E_C - E_V - 0,14 - 0,11 = 1,17 \text{ eV}$$

$$d) \quad V_{bi} = E_F^n - E_F^p = E_C - k_B T \ln \left(\frac{N_C}{N_D} \right) - k_B T \ln \left(\frac{N_V}{N_A} \right) - E_V = E_g - k_B T \left(\ln \left(\frac{N_C}{N_D} \right) + \ln \left(\frac{N_V}{N_A} \right) \right) = E_g - k_B T \ln \left(\frac{N_C N_V}{N_A N_D} \right) = E_g + k_B T \ln \left(\frac{N_A N_D}{N_C N_V} \right)$$

$$n_p = n_i^2 \quad n = N_C \quad p = N_V$$

$$e) \quad \begin{cases} \rho = +e N_D & -x_n < x < 0 \\ \rho = -e N_A & 0 < x < x_p \end{cases} \quad \frac{\partial E}{\partial x} = \frac{\rho}{\epsilon} \quad (\Rightarrow) \quad E = \int \frac{\rho}{\epsilon} dx$$

$$\begin{cases} E = \frac{1}{\epsilon} \int e N_D dx \\ E = \frac{1}{\epsilon} \int -e N_A dx \end{cases} \quad (\Rightarrow) \quad \begin{cases} E = \frac{e N_D}{\epsilon} x + C \\ E = -\frac{e N_A}{\epsilon} x + C \end{cases} \quad E(-x_n) = 0$$

$$x_n p_+ = x_p p_- \quad p_+ = N_D A$$

$$x = x_n + x_p \quad p_- = N_A A$$

$$V = - \int E dx = - \frac{\rho_+}{\epsilon} \frac{x^2}{2} + \frac{\rho_-}{\epsilon} \frac{x^2}{2}$$

